

Indian Standard

HIGH STRENGTH DEFORMED STEEL BARS AND WIRES FOR CONCRETE REINFORCEMENT (Fourth Revision)

1 SCOPE

This standard covers the requirements of deformed steel bars and wires for use as reinforcement in concrete. It covers nominal sizes from 4 mm up to and including 50 mm.

2 GRADES

Bars and wires shall be classified as Fe 415, Fe 415D, Fe 500, Fe 500D, Fe 550, Fe 550D, and Fe 600. The letter D denotes enhanced ductility properties required for seismic and earthquake-resistant designs.

Mechanical & Chemical Properties

4 CHEMICAL COMPOSITION

Clause 4.2 Ladle Analysis of Fe 500D

For Fe 500D grade deformed steel bars, the ladle analysis shall conform to the following limits:

- a) Carbon (C): Maximum 0.25 percent by mass.
- b) Sulfur (S): Maximum 0.040 percent by mass.
- c) Phosphorus (P): Maximum 0.040 percent by mass.
- d) Sulfur and Phosphorus combined: Maximum 0.080 percent by mass.
- e) Carbon Equivalent (CE): Maximum 0.42 percent for guaranteed weldability.

8 MECHANICAL PROPERTIES

Clause 8.1 Tensile and Elongation Requirements of Fe 500D

When tested in accordance with IS 1608, mechanical properties of Fe 500D shall satisfy:

- a) Minimum 0.2 percent proof stress or yield stress (R_e): 500.0 N/mm² (MPa).
- b) Minimum tensile strength (R_m): 565.0 N/mm² (not less than 1.10 times actual yield stress).
- c) Percentage elongation at gauge length 5.65·A: Not less than 16.0 percent.
- d) Total elongation at maximum force (A_g): Not less than 5.0 percent.